

RAHUL PANCHAL

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PROFESSIONAL EXPERIENCE

Upforce Tech - Full Stack Software Engineer

Jul 2025 – Present

- **Leadership & B2B Delivery:** Took architectural ownership of HireTrack ([hiretrack.in](#)), managing a 3-developer team, onboarding a dedicated B2B client, and establishing an internal DevOps tool with custom licensing for seamless on-premise Over-The-Air (OTA) updates.
- **Infrastructure & CI/CD:** Managed production deployments on VM (Nginx, SSL, Cloudflared, PM2) and established automated, near-zero downtime deployment pipelines using GitLab CI/CD and SonarQube.
- **Architecture & Performance Engineering:** Upgraded legacy infrastructure by designing a custom WebSocket server for real-time updates, architecting a schema-driven Custom-Fields engine, and virtualizing the Kanban board to slash board render times to under 10ms.
- **AI & Document Processing:** Integrated PyMuPDF4LLM into the AI parsing pipeline, heavily optimizing document extraction to reduce LLM input tokens and improve inference efficiency.
- **Internal Tooling & R&D (*free-ai-pool*):** Architected an in-house lightweight LLM load balancer and OpenAI-compatible proxy with a secure dashboard for API key management and failover routing, publishing it to npm where it achieves **2,200+ weekly downloads**.
- **Browser-Native R&D:** Creating a semantic search system for browser history utilizing MiniLM and vector embeddings, resolving critical search indexing bugs.
- **Realtime SDK (*@presence-sync/node*, *@presence-sync/react*, *@presence-sync/cli*):** Built and published a self-hosted live-cursor SDK for Next.js/Express, with piggyback, standalone and BYOS WebSocket adapters.

Upforce Tech - Software Engineering Intern

Jan 2025 – Jul 2025

- **Core Infrastructure:** Built foundational frontend and backend architecture from the ground up to support rapid platform scaling.
- **Full-Stack Development:** Developed and deployed 20+ core features for the initial phases of HireTrack using Next.js, React, Node.js, TypeScript, and MongoDB.
- **Multi-Tenant Architecture:** Architected the multi-tenant structure, enabling the super admin to seamlessly manage multiple organizations.
- **Platform Reliability:** Resolved 200+ bugs across the UI, backend, and database, resolving critical production issues to ensure platform stability.
- **Backend & Queuing:** Designed robust email APIs integrated with a mail queue system, successfully scaling automated notifications for updates.

PROJECTS

Selldocs – <https://selldocs.vercel.app> - (Demo login: *demo@selldocs.in* , *Demo@selldocs.in*)

Jun 2026 – present

- Built a scalable DRM (Digital Rights Management) tool using Next.js and TypeScript, allowing independent authors to securely deliver pre-release Advanced Reader Copies (ARCs) or any paid content in PDF easily.
- Developed a dynamic anti-piracy checkout pipeline that automatically embeds and flattens buyer-specific identifying data directly into PDF page upon download to successfully deter unauthorized sharing.
- Architected an asynchronous backend pipeline utilizing AWS Lambda and SQS to dynamically process, watermark, and flatten heavy PDF files without blocking the main application thread, ensuring highly available delivery via S3.
- Technologies used: Node.js, Next.js, MongoDB, AWS Lambda, S3, SQS, SMTP, Razorpay

React-Brai – <https://react-brai.vercel.app>

Jan 2026 – Mar 2026

- Secure, Local Intelligence interface for browsers.
- Built an experimental library to bring Edge AI capabilities directly into React applications, running machine learning models purely on the client side.
- Solved the multi-tab coordination issue with the downloaded model using worker threads.
- Successfully integrated Llama 3.2 1B and many other popular models to run entirely in the browser without backend dependencies.
- Technologies used: React.js, Next.js, WebGPU, Worker Threads

Socio – <https://socio-alpha.vercel.app> – (Demo Login: *testuser0*, *user0pass*)

Jul 2024 – Sep 2024

- Built a robust, modern social media app integrated with AI to power **personalized feeds**.
- Engineered a scalable backend architecture.
- Technologies used: React.js, Tailwind CSS, Node.js, Express.js, MongoDB, Gemini API, JWT, PM2.

EDUCATION

Bachelor of Engineering - Electronics and Communication Engineering

Government Engineering College, sector-28, Gandhinagar

May 2025

8.02 CGPA

SKILLS

Languages: TypeScript, JavaScript, C++, Python, HTML/CSS

Frontend: React.js, Next.js, Redux Toolkit, WebGPU, Tailwind CSS, Zustand, React Query(Tanstack)

Backend: Node.js, Express.js, RESTful APIs, WebSockets, GraphQL, Microservices,

Databases & Cloud: MongoDB, MySQL, AWS (Lambda, S3, SQS), GCP, Nginx, Docker, GitLab CI/CD, PM2

Developer Tools: Git, SonarQube, Linux, Postman, Cursor

HONORS & AWARDS

Best performing team at TicTacToe'23 Hackathon - DAIICT

Sept 2023

- Recognized in the top 10 teams for excellent team work, decision making and problem solving.